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Pneumatic tire

Abstract

A pneumatic tire includes inner intermediate lug grooves having a length $L1$ and outer intermediate lug grooves having a length $L2$ formed in inner and outer intermediate land portions, respectively, where lengths $L1$ and $L2$ satisfy $L1 > L2$. Distances $W1$, $W2$, $W3$, and $W4$ from a tire equator to an inner shoulder main groove, an inner center main groove, an outer center main groove, and an outer shoulder main groove, respectively, satisfy $0.58 \leq W1/W_i \leq 0.62$, $0.20 \leq W2/W_i \leq 0.24$, $0.15 \leq W3/W_o \leq 0.19$, and $0.53 \leq W4/W_o \leq 0.57$ with respect to distances W_i , W_o from the tire equator to ground contact edges. A total value of a groove width $WG1$ of the inner shoulder main groove and a groove width $WG2$ of the inner center main groove is larger than a total value of a groove width $WG3$ of the outer center main groove and a groove width $WG4$ of the outer shoulder main groove.

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Background/Summary

TECHNICAL FIELD

(1) The present technology relates to a pneumatic in which four main grooves are provided in a tread portion and particularly relates to a pneumatic tire whereby steering stability, noise performance, and drainage performance can be improved in a well-balanced manner.

BACKGROUND ART

(2) To ensure drainage performance, a pneumatic tire that employs a tread pattern in which four main grooves extending in a tire circumferential direction are formed in a tread portion and a

plurality of lug grooves are formed in land portions defined by the main grooves has been variously proposed (for example, see Japan Unexamined Patent Publication Nos. 2019-051834 A and 2019-001406 A, and Japan Patent No. 5413500 B).

(3) However, when groove areas of the main grooves and the lug grooves are increased, while drainage performance can be enhanced, there are problems in that steering stability and noise performance are deteriorated. Therefore, providing steering stability, noise performance, and drainage performance in a compatible manner at a high degree is not easy and further improvement is desired.

SUMMARY

(4) The present technology provides a pneumatic tire whereby steering stability, noise performance, and drainage performance can be improved in a well-balanced manner.

(5) A pneumatic tire according to the present technology may include a tread portion, a pair of sidewall portions, a pair of bead portions, and a mounting direction indicator. The tread portion may extend in a tire circumferential direction and have an annular shape. The pair of sidewall portions may be disposed on both sides of the tread portion. The pair of bead portions may be disposed on inner sides of the sidewall portions in a tire radial direction. The mounting direction indicator may indicate a mounting direction to a vehicle. The four main grooves extending in the tire circumferential direction may be formed in the tread portion. The four main grooves may include: an inner center main groove located on a vehicle inner side with respect to a tire equator when the pneumatic tire is mounted on the vehicle; an outer center main groove located on a vehicle outer side with respect to the tire equator when the pneumatic tire is mounted on the vehicle; an inner shoulder main groove located on the vehicle inner side with respect to the inner center main groove when the pneumatic tire is mounted on the vehicle; and an outer shoulder main groove located on the vehicle outer side with respect to the outer center main groove when the pneumatic tire is mounted on the vehicle. Five rows of land portions may be defined by the four main grooves. The five rows of the land portions may include: a center land portion defined between the inner center main groove and the outer center main groove; an inner intermediate land portion defined between the inner center main groove and the inner shoulder main groove; an outer intermediate land portion defined between the outer center main groove and the outer shoulder main groove; an inner shoulder land portion defined on the vehicle inner side with respect to the inner shoulder main groove; and an outer shoulder land portion defined on the vehicle outer side with respect to the outer shoulder main groove. While a plurality of inner intermediate lug grooves extending in a tire width direction and having one ends communicating with the inner center main groove and other ends terminating in the inner intermediate land portion may be formed at intervals in the tire circumferential direction in the inner intermediate land portion, a plurality of outer intermediate lug grooves extending in the tire width direction and having one ends communicating with the outer shoulder main groove and other ends terminating in the outer intermediate land portion may be formed at intervals in the tire circumferential direction in the outer intermediate land portion. A length $L1$ of the inner intermediate lug groove and a length $L2$ of the outer intermediate lug groove may satisfy a relationship $L1 > L2$. A distance $W1$ from the tire equator to a center position of the inner shoulder main groove, a distance $W2$ from the tire equator to a center position of the inner center main groove, a distance $W3$ from the tire equator to a center position of the outer center main groove, and a distance $W4$ from the tire equator to a center position of the outer shoulder main groove may satisfy relationships $0.58 \leq W1/W1 \leq 0.62$, $0.20 \leq W2/Wi \leq 0.24$, $0.15 \leq W3/Wo \leq 0.19$, and $0.53 \leq W4/Wo \leq 0.57$ with respect to distances Wi , Wo from the tire equator to ground contact edges on the vehicle inner side and the vehicle outer side. A total value of a groove width $WG1$ of the inner shoulder main groove and a groove width $WG2$ of the inner center main groove may be larger than a total value of a groove width $WG3$ of the outer center main groove and a groove width $WG4$ of the outer shoulder main groove.

(6) According to the present technology, in the pneumatic tire in which the mounting direction to

the vehicle is designated, W1/Wi, W2/Wi, W3/Wo, and W4/Wo are defined as described above, and the four main grooves are brought closer to the vehicle inner side when the tire is mounted on the vehicle, whereby it is possible to improve drainage performance while vehicle external noise is reduced. The total value of the groove width WG1 of the inner shoulder main and the groove width WG2 of the inner center main groove is larger than the total value of the groove width WG3 of the outer center main groove and the groove width WG4 of the outer shoulder main groove. Thus, drainage performance can be improved without deteriorating noise performance. In addition to the total value of the groove width WG1 of the inner shoulder main and the groove width WG2 of the inner center main groove being larger than the total value of the groove width WG3 of the outer center main groove and the groove width WG4 of the outer shoulder main groove, by configuring the length L1 of the inner intermediate lug groove larger than the length L2 of the outer intermediate lug groove, rigidity of the vehicle outer side part of the tread portion on which a large lateral force acts is made larger than rigidity of the vehicle inner side part of the tread portion, thus allowing improving steering stability on dry road surfaces and wet road surfaces. As a result, steering stability, noise performance, and drainage performance can be improved to a further high degree.

(7) In the present technology, in the center land portion, a plurality of center lug grooves extending in the tire width direction and having one ends communicating with the inner center main groove and other ends terminating in the center land portion are preferably formed at intervals in the tire circumferential direction. By disposing the non-through center lug grooves in the center land portion, drainage performance can be improved without impairing steering stability.

(8) In particular, the other ends of the center lug grooves preferably terminate within the center land portion without traversing the tire equator. By disposing the center lug grooves such that the center lug grooves do not go traversing the tire equator, the rigidity of the center land portion can be sufficiently ensured and drainage performance can be improved without deteriorating steering stability.

(9) In the center land portion, a plurality of center sipes extending in the tire width direction and having one ends communicating with the outer center main groove and other ends terminating within the center land portion without traversing the tire equator are preferably formed at intervals in the tire circumferential direction. By disposing the center sipes such that the center sipes do not traverse the tire equator, the rigidity of the center land portion can be sufficiently ensured and drainage performance can be improved without deteriorating noise performance or steering stability.

(10) A width Wr1 of the inner intermediate land portion and a length L1 of the inner intermediate lug groove preferably satisfy a relationship $0.6 \leq L1/Wr1 \leq 0.7$. A width Wr2 of the outer intermediate land portion and a length L2 of the outer intermediate lug groove preferably satisfy a relationship $0.4 \leq L2/Wr2 \leq 0.5$. A width Wr3 of the center land portion and a length L3 of the center lug groove preferably satisfy a relationship $0.3 \leq L3/Wr3 \leq 0.4$. By properly setting the length L1 of the inner intermediate lug groove, the length L2 of the outer intermediate lug groove, or the length L3 of the center lug groove, steering stability, noise performance, and drainage performance can be improved in a well-balanced manner.

(11) An inclination angle $\theta 1$ on an acute angle side of the inner intermediate lug groove with respect to the tire circumferential direction is preferably smaller than an inclination angle $\theta 2$ on an acute angle side of the outer intermediate lug groove with respect to the tire circumferential direction. By setting the inclination angle $\theta 1$ of the inner intermediate lug groove to be smaller than the inclination angle $\theta 2$ of the outer intermediate lug groove in this manner, the rigidity of the inner intermediate lug groove can be ensured while good drainage performance is maintained. As a result, the steering stability on dry road surfaces and the steering stability on wet road surfaces can be improved in a well-balanced manner.

(12) Groove walls on the vehicle outer sides of the inner shoulder main groove and the outer center

main groove preferably have edge portions chamfered in a zigzag shape. By providing the edge portions where the groove walls on the vehicle outer sides of the inner shoulder main groove and the outer center main groove are chamfered in the zigzag shape, an initial groove area can be ensured and drainage performance can be improved without deteriorating noise.

(13) The inner intermediate lug groove preferably has an inclined portion that is inclined such that a groove depth gradually shallows toward a terminal end side from a position 50% to 65% of the length L1 with an open end of the inner intermediate lug groove as a point of origin. The outer intermediate lug groove preferably has an inclined portion that is inclined such that a groove depth gradually shallows toward a terminal end side from a position 35% to 50% of the length L2 with an open end of the outer intermediate lug groove as a point of origin. The center lug groove preferably has an inclined portion that is inclined such that a groove depth gradually shallows toward a terminal end side from a position 25% to 40% of the length L3 with an open end of the center lug groove as a point of origin. By providing the inclined portion, the decrease in rigidity of each of the land portions can be avoided, thus allowing improving drainage performance without deteriorating steering stability.

(14) In the present technology, the ground contact region of the tread portion is specified based on the ground contact width in a tire axial direction measured when a regular load is applied and the tire is placed vertically on a plane, in a state where the tire is inflated to a regular internal pressure and mounted on a regular rim. The ground contact edge is an outermost position in the tire axial direction of the ground contact region. “Regular rim” refers to a rim defined by a standard for each tire according to a system of standards that includes standards with which tires comply, and is “standard rim” defined by Japan Automobile Tyre Manufacturers Association (JATMA), “Design Rim” defined by The Tire and Rim Association, Inc. (TRA), or “Measuring Rim” defined by European Tire and Rim Technical Organization (ETRTO), for example. “Regular internal pressure” is an air pressure corresponding to the maximum load capacity determined for each tire by each standard in a standard system including the standard on which the tire is based. “Regular load” is a load equivalent to 80% of the maximum load capacity determined for each tire by each standard in a standard system including the standard on which the tire is based.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a meridian cross-sectional view illustrating a pneumatic tire according to an embodiment of the present technology.

(2) FIG. 2 is a developed view illustrating a tread pattern of the pneumatic tire of FIG. 1.

(3) FIG. 3 is a contour diagram illustrating a contour of a meridian cross-section of a tread portion of the pneumatic tire in FIG. 1.

DETAILED DESCRIPTION

(4) Configurations of embodiments of the present technology will be described in detail below with reference to the accompanying drawings. FIGS. 1 to 3 illustrate a pneumatic tire according to an embodiment of the present technology. This pneumatic tire has a mounting direction of tire front/back that is designated when mounted on a vehicle. In FIGS. 1 to 3, “IN” refers to a vehicle inner side when the tire is mounted on a vehicle, “OUT” refers to a vehicle outer side when the tire is mounted on the vehicle, Ei refers to a ground contact edge on the vehicle inner side, Eo refers to a ground contact edge on the vehicle outer side, and TCW refers to a ground contact width.

(5) As illustrated in FIG. 1, a pneumatic tire of the present embodiment includes an annular tread portion 1 extending in the tire circumferential direction, a pair of sidewall portions 2, 2 disposed on both sides of the tread portion 1, and a pair of bead portions 3, 3 disposed on an inner side of the sidewall portions 2 in the tire radial direction. For example, a mounting direction indicator 2A

indicating the mounting direction to the vehicle is formed on at least the sidewall portion **2** on the vehicle outer side. The mounting direction indicator **2A** displays, for example, "OUTSIDE" along the tire circumferential direction on the vehicle outer side and displays, for example, "INSIDE" along the tire circumferential direction on the vehicle inner side.

(6) A carcass layer **4** is mounted between the pair of bead portions **3**, **3**. The carcass layer **4** includes a plurality of reinforcing cords extending in the tire radial direction and is folded back around a bead core **5** disposed in each of the bead portions **3** from a tire inner side to a tire outer side. A bead filler **6** having a triangular cross-sectional shape and formed of a rubber composition is disposed on the outer circumference of the bead core **5**.

(7) On the other hand, a plurality of belt layers **7** are embedded on the outer circumferential side of the carcass layer **4** in the tread portion **1**. The belt layers **7** include a plurality of reinforcing cords that are inclined with respect to the tire circumferential direction, and the reinforcing cords are disposed so as to intersect each other between the layers. In the belt layers **7**, the inclination angle of the reinforcing cords with respect to the tire circumferential direction is set to fall in a range from 10° to 40°, for example. Steel cords are preferably used as the reinforcing cords of the belt layers **7**. To improve high-speed durability, at least one belt cover layer **8** formed by arranging reinforcing cords at an angle of, for example, 5° or less with respect to the tire circumferential direction is disposed on an outer circumferential side of the belt layers **7**. Organic fiber cords such as nylon and aramid are preferably used as the reinforcing cords of the belt cover layer **8**.

(8) Note that the tire internal structure described above represents a typical example for a pneumatic tire, but the pneumatic tire is not limited thereto.

(9) As illustrated in FIG. 2, the tread portion **1** is formed with four main grooves **11** extending in the tire circumferential direction. The four main grooves **11** include an inner center main groove **11B** located on a vehicle inner side with respect to a tire equator CL when the pneumatic tire is mounted on a vehicle, an outer center main groove **11C** located on a vehicle outer side with respect to the tire equator CL when the pneumatic tire is mounted on a vehicle, an inner shoulder main groove **11A** located on the vehicle inner side with respect to the inner center main groove **11B** when the pneumatic tire is mounted on a vehicle, and an outer shoulder main groove **11D** located on the vehicle outer side with respect to the outer center main groove **11C** when the pneumatic tire is mounted on a vehicle. The main grooves **11** have groove widths ranging from 6.5 mm to 12.0 mm and groove depths ranging from 6.5 mm to 8.5 mm, and includes a wear indicator.

(10) The tread portion **1** defines five rows of land portions **12** by the four main grooves **11**. The five rows of the land portions **12** include a center land portion **12C** defined between the inner center main groove **11B** and the outer center main groove **11C**, an inner intermediate land portion **12B** defined between the inner center main groove **11B** and the inner shoulder main groove **11A**, an outer intermediate land portion **12D** defined between the outer center main groove **11C** and the outer shoulder main groove **11D**, an inner shoulder land portion **12A** defined on the vehicle inner side with respect to the inner shoulder main groove **11A**, and an outer shoulder land portion **12E** defined on the vehicle outer side with respect to the outer shoulder main groove **11D**.

(11) In the inner shoulder land portion **12A**, a plurality of inner shoulder lug grooves **13A** extending in the tire width direction and having one ends extending to the vehicle inner side with respect to the ground contact edge Ei on the vehicle inner side and the other ends terminating in the inner shoulder land portion **12A** are formed at intervals in the tire circumferential direction. Furthermore, in the inner shoulder land portion **12A**, a plurality of inner shoulder sipes **14A** extending in the tire width direction and having one ends communicating with the inner shoulder main groove **11A** and the other ends terminating in the inner shoulder land portion **12A** are formed at intervals in the tire circumferential direction.

(12) In the inner intermediate land portion **12B**, a plurality of inner intermediate lug grooves **13B** extending in the tire width direction and having one ends communicating with the inner center main groove **11B** and the other ends terminating in the inner intermediate land portion **12B** are

formed at intervals in the tire circumferential direction. In addition, on a groove wall on the vehicle outer side of the inner shoulder main groove **11A** (that is, a side wall on the vehicle inner side of the inner intermediate land portion **12B**), edge portions **16B** chamfered in a zigzag shape are formed.

(13) In the center land portion **12C**, a plurality of center lug grooves **13C** extending in the tire width direction and having one ends communicating with the inner center main groove **11B** and the other ends terminating in the center land portion **12C** are formed at intervals in the tire circumferential direction. In the center land portion **12C**, a plurality of center sipes **14C** extending in the tire width direction and having one ends communicating with the outer center main groove **11C** and the other ends terminating in the center land portion **12C** are formed at intervals in the tire circumferential direction.

(14) In the outer intermediate land portion **12D**, a plurality of outer intermediate lug grooves **13D** extending in the tire width direction and having one ends communicating with the outer shoulder main groove **11D** and the other ends terminating in the outer intermediate land portion **12D** are formed at intervals in the tire circumferential direction. In addition, on a groove wall on the vehicle outer side of the outer center main groove **11C** (that is, a side wall on the vehicle inner side of the outer intermediate land portion **12D**), edge portions **16D** chamfered in a zigzag shape are formed.

(15) In the outer shoulder land portion **12E**, a plurality of outer shoulder lug grooves **13E** extending in the tire width direction and having one ends extending to the vehicle outer side with respect to the ground contact edge **Eo** on the vehicle outer side and the other ends terminating in the outer shoulder land portion **12E** are formed at intervals in the tire circumferential direction. In the outer shoulder land portion **12E**, a circumferential auxiliary groove **15E** extending in the tire circumferential direction and mutually connecting the outer shoulder lug grooves **13E** is formed. In the outer shoulder land portion **12E**, a plurality of outer shoulder sipes **14D** extending in the tire width direction and having one ends communicating with the outer shoulder main groove **11D** and the other ends communicating with the circumferential auxiliary groove **15E** are formed at intervals in the tire circumferential direction.

(16) In the pneumatic tire, the length **L1** of the inner intermediate lug groove **13B** and the length **L2** of the outer intermediate lug groove **13D** satisfy the relationship $L1 > L2$. The lengths **L1**, **L2** are measured in the tire width direction. By configuring the length **L1** of the inner intermediate lug groove **13B** larger than the length **L2** of the outer intermediate lug groove **13D** in this way, rigidity of the vehicle outer side part of the tread portion **1** (the outer intermediate land portion **12D**) on which a large lateral force acts is made larger than rigidity of the vehicle inner side part of the tread portion **1** (the inner intermediate land portion **12B**), thus allowing improving the steering stability on dry road surfaces and wet road surfaces.

(17) In the pneumatic tire, a distance **W1** from the tire equator **CL** to the center position of the inner shoulder main groove **11A**, a distance **W2** from the tire equator **CL** to the center position of the inner center main groove **11B**, a distance **W3** from the tire equator **CL** to the center position of the outer center main groove **11C**, and a distance **W4** from the tire equator **CL** to the center position of the outer shoulder main groove **11D** satisfy the relationships $0.58 \leq W1/Wi \leq 0.62$, $0.20 \leq W2/Wi \leq 0.24$, $0.15 \leq W3/Wo \leq 0.19$, and $0.53 \leq W4/Wo \leq 0.57$ with respect to distances **Wi**, **Wo** from the tire equator **CL** to the ground contact edges **Ei**, **Eo** on the vehicle inner side and the vehicle outer side. The respective distances **Wi** and **Wo** from the tire equator **CL** to the ground contact edges **Ei**, **Eo** on the vehicle inner side and the vehicle outer side are equivalent to $TCW/2$. The center positions of the inner shoulder main groove **11A**, the inner center main groove **11B**, the outer center main groove **11C**, and the outer shoulder main groove **11D** are center positions in a groove width direction and center positions of the groove portions excluding the chamfered portions.

(18) In the pneumatic tire in which the mounting direction to the vehicle is designated, $W1/Wi$, $W2/Wi$, $W3/Wo$, and $W4/Wo$ are defined as described above, and the four main grooves are

brought closer to the vehicle inner side when the tire is mounted on a vehicle, whereby it is possible to improve drainage performance while vehicle external noise is reduced. Here, when the center positions of the inner shoulder main groove **11A**, the inner center main groove **11B**, the outer center main groove **11C**, and the outer shoulder main groove **11D** are out of the ranges, it is difficult to provide noise performance and drainage performance in a compatible manner.

(19) In the pneumatic tire, the total value of the groove width **WG1** of the inner shoulder main groove **11A** and the groove width **WG2** of the inner center main groove **11B** (**WG1+WG2**) is set to be larger than the total value of the groove width **WG3** of the outer center main groove **11C** and the groove width **WG4** of the outer shoulder main groove **11D** (**WG3+WG4**). Thus, drainage performance can be improved without deteriorating noise performance. Configuring the total value of the groove width **WG1** of the inner shoulder main groove **1A** and the groove width **WG2** of the inner center main groove **11B** to be larger than the total value of the groove width **WG3** of the outer center main groove **11C** and the groove width **WG4** of the outer shoulder main groove **11D** allows increasing the rigidity of the vehicle outer side part of the tread portion **1** where a large lateral force acts more than the rigidity of the vehicle inner side part of the tread portion **1**, thus allowing improving the steering stability on dry road surfaces and wet road surfaces. In particular, the relationship **WG1>WG2>WG3>WG4** is preferably satisfied.

(20) In the center land portion **12C** of the pneumatic tire, the plurality of center lug grooves **13C** extending in the tire width direction and having one ends communicating with the inner center main groove **11B** and the other ends terminating in the center land portion **12C** are preferably formed at intervals in the tire circumferential direction. By disposing the non-through center lug grooves **13C** in the center land portion **12C**, drainage performance can be improved without impairing steering stability.

(21) When the center lug grooves **13C** are provided, the other ends of the center lug grooves **13C** preferably terminate within the center land portion **12C** without traversing the tire equator **CL**. By disposing the center lug grooves **13C** such that the center lug grooves **13C** do not traverse the tire equator **CL**, the rigidity of the center land portion **12C** can be sufficiently ensured, and drainage performance can be improved without deteriorating steering stability.

(22) In the center land portion **12C** of the pneumatic tire, the plurality of center sipes **14C** extending in the tire width direction and having one ends communicating with the outer center main groove **11C** and the other ends terminating within the center land portion **12C** without traversing the tire equator **CL** are preferably formed at intervals in the tire circumferential direction. By disposing the center sipes **14C** such that the center sipes **14C** do not traverse the tire equator **CL**, the rigidity of the center land portion **12C** can be sufficiently ensured and drainage performance can be improved without deteriorating noise performance or steering stability.

(23) In the pneumatic tire, a width **Wr1** of the inner intermediate land portion **12B** and the length **L1** of the inner intermediate lug groove **13B** preferably satisfy the relationship $0.6 \leq L1/Wr1 \leq 0.7$. By properly setting the length **L1** of the inner intermediate lug groove **13B**, steering stability, noise performance, and drainage performance can be improved in a well-balanced manner. Here, the value of **L1/Wr1** being less than 0.6 decreases the effect of improving drainage performance and whereas the value being larger than 0.7 decreases the effect of improving steering stability due to the decrease in rigidity of the inner intermediate land portion **12B** and decreases the effect of improving noise performance.

(24) In the pneumatic tire, a width **Wr2** of the outer intermediate land portion **12D** and the length **L2** of the outer intermediate lug groove **13D** preferably satisfy the relationship $0.4 \leq L2/Wr2 \leq 0.5$. By properly setting the length **L2** of the outer intermediate lug groove **13D**, steering stability, noise performance, and drainage performance can be improved in a well-balanced manner. Here, the value of **L2/Wr2** being less than 0.4 decreases the effect of improving drainage performance and whereas the value being larger than 0.5 decreases the effect of improving steering stability due to the decrease in rigidity of the outer intermediate land portion **12D** and decreases the effect of

improving noise performance.

(25) In the pneumatic tire, a width $Wr3$ of the center land portion **12C** and a length $L3$ of the center lug groove **13C** preferably satisfy the relationship $0.3 \leq L3/Wr3 \leq 0.4$. By properly setting the length $L3$ of the center lug groove **13C**, steering stability, noise performance, and drainage performance can be improved in a well-balanced manner. Here, the value of $L3/Wr3$ being less than 0.3 decreases the effect of improving drainage performance and whereas the value being larger than 0.4 decreases the effect of improving steering stability due to the decrease in rigidity of the center land portion **12C** and decreases the effect of improving noise performance.

(26) In the pneumatic tire, an inclination angle $\theta 1$ on the acute angle side of the inner intermediate lug groove **13B** with respect to the tire circumferential direction is preferably set to be smaller than an inclination angle $\theta 2$ on the acute angle side of the outer intermediate lug groove **13D** with respect to the tire circumferential direction. The inclination angle $\theta 1$ is the angle formed by a straight line connecting the center position of the opening end of the inner intermediate lug groove **13B** and the center position of the terminal end thereof with respect to the tire circumferential direction, and the inclination angle $\theta 2$ is the angle formed by a straight line connecting the center position of the opening end of the outer intermediate lug groove **13D** and the center position of the terminal end thereof with respect to the tire circumferential direction. By setting the inclination angle $\theta 1$ of the inner intermediate lug groove **13B** to be smaller than the inclination angle $\theta 2$ of the outer intermediate lug groove **13D** in this manner, the rigidity of the inner intermediate lug groove **13B** can be ensured while maintaining good drainage performance. As a result, the steering stability on dry road surfaces and the steering stability on wet road surfaces can be improved in a well-balanced manner. In particular, it is preferable that the inclination angle $\theta 1$ of the inner intermediate lug groove **13B** is set in a range from 65° to 71° and the inclination angle $\theta 2$ of the outer intermediate lug groove **13D** is set in a range from 72° to 78°.

(27) In the pneumatic tire, the edge portions **16B**, **16D** chamfered in a zigzag shape are preferably formed on the groove walls on the vehicle outer sides of the inner shoulder main groove **11A** and the outer center main groove **1C**, respectively. By providing the edge portions **16B**, **16D** where the groove walls on the vehicle outer sides of the inner shoulder main groove **11A** and the outer center main groove **1C** are chamfered in the zigzag shape, the initial groove area can be ensured and drainage performance can be improved without deteriorating noise.

(28) In the pneumatic tire, as illustrated in FIG. 3, the inner intermediate lug groove **13B**, the center lug groove **13C**, and the outer intermediate lug groove **13D** have inclined portions **17B**, **17C**, **17D** inclined such that the groove depths gradually decrease from the intermediate position toward the terminal end side in the longitudinal direction. More specifically, the groove depth of the inner intermediate lug groove **13B** gradually shallows toward the terminal end side from a position away from the open end thereof as a point of origin by a distance $L1'$ equivalent to 50% to 650% of the length $L1$. The groove depth of the outer intermediate lug groove **13D** gradually shallows toward the terminal end side from a position away from the open end thereof as a point of origin by a distance $L2'$ equivalent to 350% to 50% of the length $L2$. The groove depth of the center lug groove **13C** gradually shallows toward the terminal end side from a position away from the open end thereof as a point of origin by a distance $L3'$ equivalent to 25% to 40% of the length $L3$. By providing the inclined portions **17B**, **17C**, **17D**, the decrease in rigidity of the inner intermediate lug groove **13B**, the center lug groove **13C**, or the outer intermediate lug groove **13D** can be avoided, thus allowing improving drainage performance without deteriorating steering stability. Here, when the distances $L1'$, $L2'$, $L3'$ are excessively small, the effect of improving drainage performance decreases, and when the distances $L1'$, $L2'$, $L3'$ are excessively large, the effect of improving steering stability decreases.

EXAMPLES

(29) Tires of Conventional Example, Comparative Examples 1 to 3, and Examples 1 to 8 having tire sizes of 215/55R17 94V that were pneumatic tires including tread portions, a pair of sidewall

portions, a pair of bead portions, and mounting direction indicators indicating a mounting direction to a vehicle and in which, as illustrated in FIG. 2, four main grooves extending in the tire circumferential directions were formed in the tread portions, five rows of land portions were defined by the four main grooves, and lug grooves and sipes were formed in the land portions were manufactured.

(30) In Conventional Example, Comparative Examples 1 to 3, and Examples 1 to 8, the distances W_i , W_o from the tire equator to the ground contact edges, the length $L1$ of the inner intermediate lug groove, the length $L2$ of the outer intermediate lug groove, the length $L3$ of the center lug groove, the distance $W1$ from the tire equator to the center position of the inner shoulder main groove, the distance $W2$ from the tire equator to the center position of the inner center main groove, the distance $W3$ from the tire equator to the center position of the outer center main groove, the distance $W4$ from the tire equator to the center position of the outer shoulder main groove, $W1/W_i$, $W2/W_i$, $W3/W_o$, $W4/W_o$, the total value of the groove width $WG1$ of the inner shoulder main groove and the groove width $WG2$ of the inner center main groove ($WG1+WG2$), the total value of the groove width $WG3$ of the outer center main groove and the groove width $WG4$ of the outer shoulder main groove ($WG3+WG4$), the termination position of a non-through center lug groove, presence of the non-through center sipe, the width $Wr1$ of the inner intermediate land portion, the width $Wr2$ of the outer intermediate land portion, the width $Wr3$ of the center land portion, $L1/Wr1$, $L2/Wr2$, $L3/Wr3$, the inclination angle $\theta1$ of the inner intermediate lug groove, the inclination angle $\theta2$ of the outer intermediate lug groove, presence of the chamfering in a zigzag shape on the groove walls on the vehicle outer sides of the inner shoulder main groove and the outer center main groove, the start position $L1'$ of the inclined portion of the inner intermediate lug groove, the start position $L2'$ of the inclined portion of the outer intermediate lug groove, the start position $L3'$ of the inclined portion of the center lug groove, $L1'/L1$, $L2'/L2$, and $L3'/L3$ were set as in Table 1 and Table 2. Regarding the terminal end position of the non-through center lug groove, the case where the terminal end was on the tire equator was evaluated as "A" and the case where the terminal end was at a position not traversing the tire equator was evaluated as "B."

(31) These test tires were evaluated for the steering stability on dry road surfaces, the steering stability on wet road surfaces, and noise performance by the following test methods, and the results are also given in Table 1 and Table 2.

(32) Steering stability on dry road surfaces: Each test tire was mounted on a wheel with a rim size of 17×7J and mounted on a passenger vehicle with an engine displacement of 2500 cc, air pressure (F/R) after warm-up was set to 230 kPa/220 kPa, and sensory evaluation during traveling on dry road surfaces was conducted by a test driver. Evaluation results are expressed as index values with the value of Conventional Example being defined as 100. Larger index values indicate superior steering stability on dry road surfaces.

(33) Steering stability on wet road surfaces: Each test tire was mounted on a wheel with a rim size of 17×7 J and mounted on a passenger vehicle with an engine displacement of 2500 cc, air pressure (F/R) after warm-up was set to 230 kPa/220 kPa, and sensory evaluation during traveling on wet road surfaces was conducted by a test driver. Evaluation results are expressed as index values with the value of Conventional Example being defined as 100. Larger index values indicate superior steering stability on wet road surfaces.

(34) Noise performance: Each test tire was mounted on a wheel with a rim size of 17×7 J and mounted on a passenger vehicle with an engine displacement of 2500 cc, air pressure (F/R) after warm-up was set to 230 kPa/220 kPa, and a measurement test for vehicle external noise compliant with ECE R117 was conducted to measure the vehicle external noise. Evaluation results are expressed as index values, using the reciprocals of the measurement values, with the value of Conventional Example being defined as 100. Larger index values mean lower vehicle external noise and better noise performance.

(35) TABLE-US-00001 TABLE 1-1 Conventional Comparative Comparative Example Example 1

Example 2 Distances Wi, Wo to ground contact 81 81 81 edges (mm) Length of lug L1 (mm) 6.4 6.4 15 groove L2 (mm) 18 18 11 L3 (mm) 12.3 12.3 12.3 Center W1 (mm) 47 48 46 position of W2 (mm) 15.8 16.8 14.8 main groove W3 (mm) 15.8 14.8 16.8 W4 (mm) 47 46 48 W1/Wi 0.58 0.59 0.57 W2/Wi 0.20 0.21 0.18 W3/Wo 0.20 0.18 0.21 W4/Wo 0.58 0.57 0.59 Groove width of WG1 + WG2 (mm) 14.9 15.2 15.2 main groove WG3 + WG4 (mm) 14.9 14.6 14.6 Termination position of non-through A A A center lug groove Presence of non-through center sipe No No No Land portion Wr1 (mm) 24 24 23.7 width Wr2 (mm) 24 24 24.3 Wr3 (mm) 24.6 24.6 24.6 L1/Wr1 0.27 0.27 0.63 L2/Wr2 0.75 0.75 0.45 L3/Wr3 0.50 0.50 0.50 Inclination θ_1 (°) 54 54 54 angle of lug θ_2 (°) 54 54 54 groove Presence of Inner shoulder main No No No chamfering in groove zigzag shape Outer center main Yes Yes Yes groove Start position L1' (mm) — — — of inclined L2' (mm) 2.5 2.5 2.5 portion of lug L3' (mm) 2.5 2.5 2.5 groove L1'/L1 — — — L2'/L2 0.14 0.23 0.23 L3'/L3 0.20 0.20 0.20 Steering stability on dry road 100 100 100 surfaces (index value) Steering stability on wet road 100 100 99 surfaces (index value) Noise performance (index value) 100 102 99

(36) TABLE-US-00002 TABLE 1-2 Comparative Exam- Exam- Example 3 ple 1 ple 2 Distances Wi, Wo to ground contact 81 81 81 edges (mm) Length of L1 (mm) 15 15 15 lug groove L2 (mm) 11 11 11 L3 (mm) 12.3 12.3 7 Center W1 (mm) 48 48 48 position of W2 (mm) 16.8 16.8 16.8 main groove W3 (mm) 14.8 14.8 14.8 W4 (mm) 46 46 46 W1/Wi 0.59 0.59 0.59 W2/Wi 0.21 0.21 0.21 W3/Wo 0.18 0.18 0.18 W4/Wo 0.57 0.57 0.57 Groove width WG1 + WG2 (mm) 14.6 15.2 15.2 of main groove WG3 + WG4 (mm) 15.2 14.6 14.6 Termination position of non-through A A B center lug groove Presence of non-through center sipe No No Yes Land portion Wr1 (mm) 24.3 24 24 width Wr2 (mm) 23.7 24 24 Wr3 (mm) 24.6 24.6 24 L1/Wr1 0.62 0.63 0.63 L2/Wr2 0.46 0.46 0.46 L3/Wr3 0.50 0.50 0.29 Inclination θ_1 (°) 54 54 54 angle of lug θ_2 (°) 54 54 54 groove Presence of Inner shoulder No No No chamfering in main groove zigzag shape Outer center Yes Yes Yes main groove Start position L1' (mm) — — — of inclined L2' (mm) 2.5 2.5 2.5 portion of lug L3' (mm) 2.5 2.5 2.5 groove L1'/L1 — — — L2'/L2 0.23 0.23 0.23 L3'/L3 0.20 0.20 0.36 Steering stability on dry road 99 100 101 surfaces (index value) Steering stability on wet road 99 102 102 surfaces (index value) Noise performance (index value) 99 102 103

(37) TABLE-US-00003 TABLE 2-1 Example 3 Example 4 Example 5 Distances Wi, Wo to ground contact edges (mm) 81 81 81 Length of lug groove L1 (mm) 15 15 15 L2 (mm) 11 11 11 L3 (mm) 8.3 8.3 8.3 Center position of main W1 (mm) 48 48 48 groove W2 (mm) 16.8 16.8 16.8 W3 (mm) 14.8 14.8 14.8 W4 (mm) 46 46 46 W1/Wi 0.59 0.59 0.59 W2/Wi 0.21 0.21 0.21 W3/Wo 0.18 0.18 0.18 W4/Wo 0.57 0.57 0.57 Groove width of main WG1 + WG2 (mm) 15.2 15.2 15.2 groove WG3 + WG4 (mm) 14.6 14.6 14.6 Termination position of non-through center lug groove B B B Presence of non-through center sipe Yes Yes Yes Land portion width Wr1 (mm) 24 24 24 Wr2 (mm) 24 24 24 Wr3 (mm) 24 24 24 L1/Wr1 0.63 0.63 0.63 L2/Wr2 0.46 0.46 0.46 L3/Wr3 0.35 0.35 0.35 Inclination angle of lug θ_1 (°) 54 65 65 groove θ_2 (°) 54 72 72 Presence of chamfering Inner shoulder main groove No No Yes in zigzag shape Outer center main groove Yes Yes Yes Start position of inclined L1' (mm) — — — portion of lug groove L2' (mm) 2.5 2.5 2.5 L3' (mm) 2.5 2.5 2.5 L1'/L1 — — — L2'/L2 0.23 0.23 0.23 L3'/L3 0.30 0.30 0.30 Steering stability on dry road surfaces (index value) 101 102 102 Steering stability on wet road surfaces (index value) 103 103 105 Noise performance (index value) 103 104 104

(38) TABLE-US-00004 TABLE 2-2 Example 6 Example 7 Example 8 Distances Wi, Wo to ground contact edges (mm) 81 81 81 Length of lug groove L1 (mm) 15 15 15 L2 (mm) 11 11 11 L3 (mm) 8.3 8.3 8.3 Center position of main W1 (mm) 48 48 48 groove W2 (mm) 16.8 16.8 16.8 W3 (mm) 14.8 14.8 14.8 W4 (mm) 46 46 46 W1/Wi 0.59 0.59 0.59 W2/Wi 0.21 0.21 0.21 W3/Wo 0.18 0.18 0.18 W4/Wo 0.57 0.57 0.57 Groove width of main WG1 + WG2 (mm) 15.2 15.2 15.2 groove WG3 + WG4 (mm) 14.6 14.6 14.6 Termination position of non-through center lug groove B B B Presence of non-through center sipe Yes Yes Yes Land portion width Wr1 (mm) 24 24 24 Wr2 (mm) 24 24 24 Wr3 (mm) 24 24 24 L1/Wr1 0.63 0.63 0.63 L2/Wr2 0.46 0.46 0.46 L3/Wr3 0.35

0.35	0.35	Inclination angle of lug $\theta 1$ (°)	65	65	65	groove $\theta 2$ (°)	72	72	72	Presence of chamfering	
Inner shoulder main groove	Yes	Yes	Yes	Yes	in zigzag shape	Outer center main groove	Yes	Yes	Yes		
Start position of inclined L1' (mm)	9	9	9	portion of lug groove	L2' (mm)	2.5	5.3	5.3	L3' (mm)	2.5	
2.5	2.5	L1'/L1	0.6	0.6	0.6	L2'/L2	0.23	0.48	0.48	L3'/L3	0.30
0.30	0.30	0.30	Steering stability on dry road surfaces (index value)	103	104	105	Steering stability on wet road surfaces (index value)	105	105	105	
Noise performance (index value)	104	104	105								

(39) As can be seen from Table 1 and Table 2, the tires of Examples 1 to 8 have been totally good in the steering stability on dry road surfaces, the steering stability on wet road surfaces, and noise performance in comparison with Conventional Example. On the other hand, in Comparative Example 1, since the length L1 of the inner intermediate lug groove was smaller than the length L2 of the outer intermediate lug groove, the effect of improving the steering stability on dry road surfaces and wet road surfaces was insufficient. In Comparative Example 2, since the four main grooves were brought closer to the vehicle outer side when the tire was mounted on the vehicle, not only noise performance was deteriorated but also the steering stability on wet road surfaces was deteriorated. In Comparative Example 3, since the total value of the groove width WG1 of the inner shoulder main groove and the groove width WG2 of the inner center main groove (WG1+WG2) was smaller than the total value of the groove width WG3 of the outer center main groove and the groove width WG4 of the outer shoulder main groove (WG3+WG4), not only noise performance was deteriorated but also the steering stability on dry road surfaces and wet road surfaces was deteriorated.

Claims

1. A pneumatic tire, comprising: a tread portion extending in a tire circumferential direction and having an annular shape; a pair of sidewall portions disposed on both sides of the tread portion; a pair of bead portions disposed on inner sides of the sidewall portions in a tire radial direction; and a mounting direction indicator indicating a mounting direction to a vehicle, four main grooves extending in the tire circumferential direction being formed in the tread portion, the four main grooves comprising: an inner center main groove located on a vehicle inner side with respect to a tire equator when the pneumatic tire is mounted on the vehicle; an outer center main groove located on a vehicle outer side with respect to the tire equator when the pneumatic tire is mounted on the vehicle; an inner shoulder main groove located on the vehicle inner side with respect to the inner center main groove when the pneumatic tire is mounted on the vehicle; and an outer shoulder main groove located on the vehicle outer side with respect to the outer center main groove when the pneumatic tire is mounted on the vehicle, five rows of land portions being defined by the four main grooves, the five rows of the land portions comprising: a center land portion defined between the inner center main groove and the outer center main groove; an inner intermediate land portion defined between the inner center main groove and the inner shoulder main groove; an outer intermediate land portion defined between the outer center main groove and the outer shoulder main groove; an inner shoulder land portion defined on the vehicle inner side with respect to the inner shoulder main groove; and an outer shoulder land portion defined on the vehicle outer side with respect to the outer shoulder main groove, while a plurality of inner intermediate lug grooves extending in a tire width direction and having one ends communicating with the inner center main groove and other ends terminating in the inner intermediate land portion are formed at intervals in the tire circumferential direction in the inner intermediate land portion, a plurality of outer intermediate lug grooves extending in the tire width direction and having one ends communicating with the outer shoulder main groove and other ends terminating in the outer intermediate land portion are formed at intervals in the tire circumferential direction in the outer intermediate land portion, a length L1 of the inner intermediate lug groove and a length L2 of the outer intermediate lug groove satisfy a relationship $L1 > L2$, a distance W1 from the tire equator to a center position of

the inner shoulder main groove, a distance W2 from the tire equator to a center position of the inner center main groove, a distance W3 from the tire equator to a center position of the outer center main groove, and a distance W4 from the tire equator to a center position of the outer shoulder main groove satisfying relationships $0.58 \leq W1/Wi \leq 0.62$, $0.20 \leq W2/Wi \leq 0.24$, $0.15 \leq W3/Wo \leq 0.19$, and $0.53 \leq W4/Wo \leq 0.57$ with respect to distances Wi, Wo from the tire equator to ground contact edges on the vehicle inner side and the vehicle outer side, and a total value of a groove width WG1 of the inner shoulder main groove and a groove width WG2 of the inner center main groove is larger than a total value of a groove width WG3 of the outer center main groove and a groove width WG4 of the outer shoulder main groove.

2. The pneumatic tire according to claim 1, wherein in the center land portion, a plurality of center sipes extending in the tire width direction and having one ends communicating with the outer center main groove and other ends terminating within the center land portion without traversing the tire equator are formed at intervals in the tire circumferential direction.

3. The pneumatic tire according to claim 1, wherein a width Wr1 of the inner intermediate land portion and a length L1 of the inner intermediate lug groove satisfy a relationship $0.6 \leq L1/Wr1 \leq 0.7$.

4. The pneumatic tire according to claim 1, wherein a width Wr2 of the outer intermediate land portion and a length L2 of the outer intermediate lug groove satisfy a relationship $0.4 \leq L2/Wr2 \leq 0.5$.

5. The pneumatic tire according to claim 1, wherein an inclination angle $\theta 1$ on an acute angle side of the inner intermediate lug groove with respect to the tire circumferential direction is smaller than an inclination angle $\theta 2$ on an acute angle side of the outer intermediate lug groove with respect to the tire circumferential direction.

6. The pneumatic tire according to claim 1, wherein groove walls on the vehicle outer sides of the inner shoulder main groove and the outer center main groove have edge portions chamfered in a zigzag shape.

7. The pneumatic tire according to claim 1, wherein the inner intermediate lug groove has an inclined portion that is inclined such that a groove depth gradually shallows toward a terminal end side from a position 50% to 65% of the length L1 with an open end of the inner intermediate lug groove as a point of origin.

8. The pneumatic tire according to claim 1, wherein the outer intermediate lug groove has an inclined portion that is inclined such that a groove depth gradually shallows toward a terminal end side from a position 35% to 50% of the length L2 with an open end of the outer intermediate lug groove as a point of origin.

9. The pneumatic tire according to claim 1, wherein the inner and outer intermediate land portions are formed without any sipes.

10. The pneumatic tire according to claim 1, wherein the plurality of inner intermediate lug grooves are the only lug grooves formed in the inner intermediate land portion and the plurality of outer intermediate lug grooves are the only lug grooves formed in the outer intermediate land portion.

11. The pneumatic tire according to claim 10, wherein the inner and outer intermediate land portions are formed without any sipes.

12. The pneumatic tire according to claim 1, wherein in the center land portion, a plurality of center lug grooves extending in the tire width direction and having one ends communicating with the inner center main groove and other ends terminating in the center land portion are formed at intervals in the tire circumferential direction.

13. The pneumatic tire according to claim 12, wherein the center lug groove has an inclined portion that is inclined such that a groove depth gradually shallows toward a terminal end side from a position 25% to 40% of the length L3 with an open end of the center lug groove as a point of origin.

14. The pneumatic tire according to claim 12, wherein a width Wr3 of the center land portion and a length L3 of the center lug groove satisfy a relationship $0.3 \leq L3/Wr3 \leq 0.4$.

15. The pneumatic tire according to claim 12, wherein the other ends of the center lug grooves

terminate within the center land portion without traversing the tire equator.

16. The pneumatic tire according to claim 15, wherein a width $Wr3$ of the center land portion and a length $L3$ of the center lug groove satisfy a relationship $0.3 \leq L3/Wr3 \leq 0.4$.

17. The pneumatic tire according to claim 15, wherein in the center land portion, a plurality of center sipes extending in the tire width direction and having one ends communicating with the outer center main groove and other ends terminating within the center land portion without traversing the tire equator are formed at intervals in the tire circumferential direction.

18. The pneumatic tire according to claim 17, wherein a width $Wr1$ of the inner intermediate land portion and a length $L1$ of the inner intermediate lug groove satisfy a relationship $0.6 \leq L1/Wr1 \leq 0.7$.

19. The pneumatic tire according to claim 18, wherein a width $Wr2$ of the outer intermediate land portion and a length $L2$ of the outer intermediate lug groove satisfy a relationship $0.4 \leq L2/Wr2 \leq 0.5$.

20. The pneumatic tire according to claim 19, wherein an inclination angle $\theta1$ on an acute angle side of the inner intermediate lug groove with respect to the tire circumferential direction is smaller than an inclination angle $\theta2$ on an acute angle side of the outer intermediate lug groove with respect to the tire circumferential direction.

21. The pneumatic tire according to claim 20, wherein groove walls on the vehicle outer sides of the inner shoulder main groove and the outer center main groove have edge portions chamfered in a zigzag shape.

22. The pneumatic tire according to claim 21, wherein the inner intermediate lug groove has an inclined portion that is inclined such that a groove depth gradually shallows toward a terminal end side from a position 50% to 65% of the length $L1$ with an open end of the inner intermediate lug groove as a point of origin.

23. The pneumatic tire according to claim 22, wherein the outer intermediate lug groove has an inclined portion that is inclined such that a groove depth gradually shallows toward a terminal end side from a position 35% to 50% of the length $L2$ with an open end of the outer intermediate lug groove as a point of origin.
